

Freight Rate Prediction Machine Learning Assessment

Candidate Report

Objective Build and validate a machine-learning model that predicts freight rates from historical load data, then generate predictions for 12,000 held-out loads and a fixed December 2025 scenario.

1. Executive Summary

The solution used a time-aware validation strategy because the labeled development data ends on 31 October 2025, while the final validation set covers November and December 2025. This setup more closely reflects the real forecasting task than a random split.

Data cleaning addressed negative and missing weights, missing market-index values, and date parsing. Date-derived features and an origin-destination route feature were added. Several regression approaches were explored, with CatBoost selected for the final model because it handled mixed numerical and categorical freight data effectively and produced the strongest validation performance in the completed experiments.

2. Data Overview

Dataset	Rows	Purpose
train-test.csv	48,000	Labeled development data
validation.csv	12,000	Final freight-rate predictions
december-chart-inputs.csv	31	Fixed December chart scenario

The target variable is `posted_rate`. Predictors include pickup and delivery locations, coordinates, distance, equipment type, weight, date, `market_index`, and `quote_signal`.

3. Data Quality and Cleaning

- 300 missing values were identified in `weight` in the development data.
- 374 missing values were identified in `market_index` in the development data.
- 292 negative weight values were found and treated as sign errors by taking their absolute values.
- The date field was converted from text to a datetime type before feature engineering and splitting.

- Missing numerical values were filled using medians learned from the training portion only during validation to avoid leakage.

4. Feature Engineering

The following features were derived from date:

- month
- day_of_month
- day_of_week

A route feature was created by concatenating pickup and delivery (for example, Dallas_Chicago). This lets the model learn lane-specific pricing behavior that may not be captured by origin and destination independently.

5. Validation Strategy

A random train/test split was not used for final evaluation because it would mix earlier and later dates. Instead, the model was evaluated on future periods relative to the training window. Rolling time-based backtests were used to check stability across multiple two-month validation periods.

Validation period	MAE (\$)	RMSE (\$)	R ²
May-Jun 2025	141.85	644.88	0.8262
Jul-Aug 2025	149.83	627.33	0.8229
Sep-Oct 2025	113.24	632.96	0.8280

Why this split? The assessment validation data represents later months than the development data. A time-based split therefore tests the model under conditions that better match deployment.

6. Model Development

The modeling workflow compared simpler baselines with tree-based methods. Linear Regression, Random Forest, Gradient Boosting, and CatBoost were tested during development. CatBoost was selected for final predictions because it performed best in the completed validation experiments and naturally handles categorical features such as pickup, delivery, equipment, and route.

7. Final Training and Prediction

After validation, the selected CatBoost model was retrained using all 48,000 labeled rows and used to generate predicted_rate for every load in validation.csv. The final submission file contains exactly two columns: load_id and predicted_rate, with 12,000 rows.

For the fixed December scenario, a separate CatBoost model used only the predictors available in december-chart-inputs.csv, together with date-derived features and route.

8. December 2025 Prediction Chart

The provided scorer validates one prediction for each day from 1-31 December 2025 using fixed inputs: Lexington to Fort Wayne, 360 miles, Dry Van, and 32,000 lb. Only the date changes.

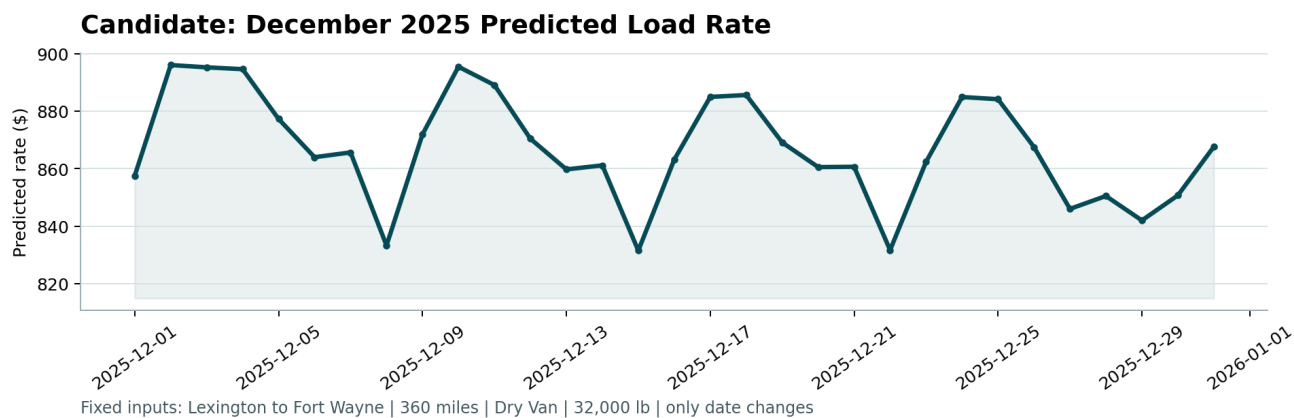


Figure 1. Fixed December 2025 predicted load rate.

9. Output Validation

The provided score.py script was used to validate the structure of the final prediction files. It checks that the validation output contains exactly 12,000 unique load IDs with positive predicted rates, validates the 31-row December file, and generates the December chart. Final hidden validation metrics are calculated by Spotter after submission.

10. Conclusion

The final workflow combines practical data-quality handling, time-aware validation, lightweight feature engineering, and CatBoost regression. The time-based backtests show broadly consistent explanatory performance across successive periods, while the final pipeline produces all required prediction outputs in the format expected by the assessment scorer.